Cloud computing fundamentals

p.vijay krishna

1.

cloud computing
computing sewices delvired over Internet

means

Defination: Sewers storage d atabases networking software available online.

has

charecteristics:
on demand acess Board network access resources polling elasticity measure service

delivers

service delivory
Resource supplied when needed, usage can be measured.

Includes

on demand network access: users request resources themsleves and access the from phones, laptop or pc's

Includes

polling+clasticity
shared provider resource capcity grows or shrinks with demand

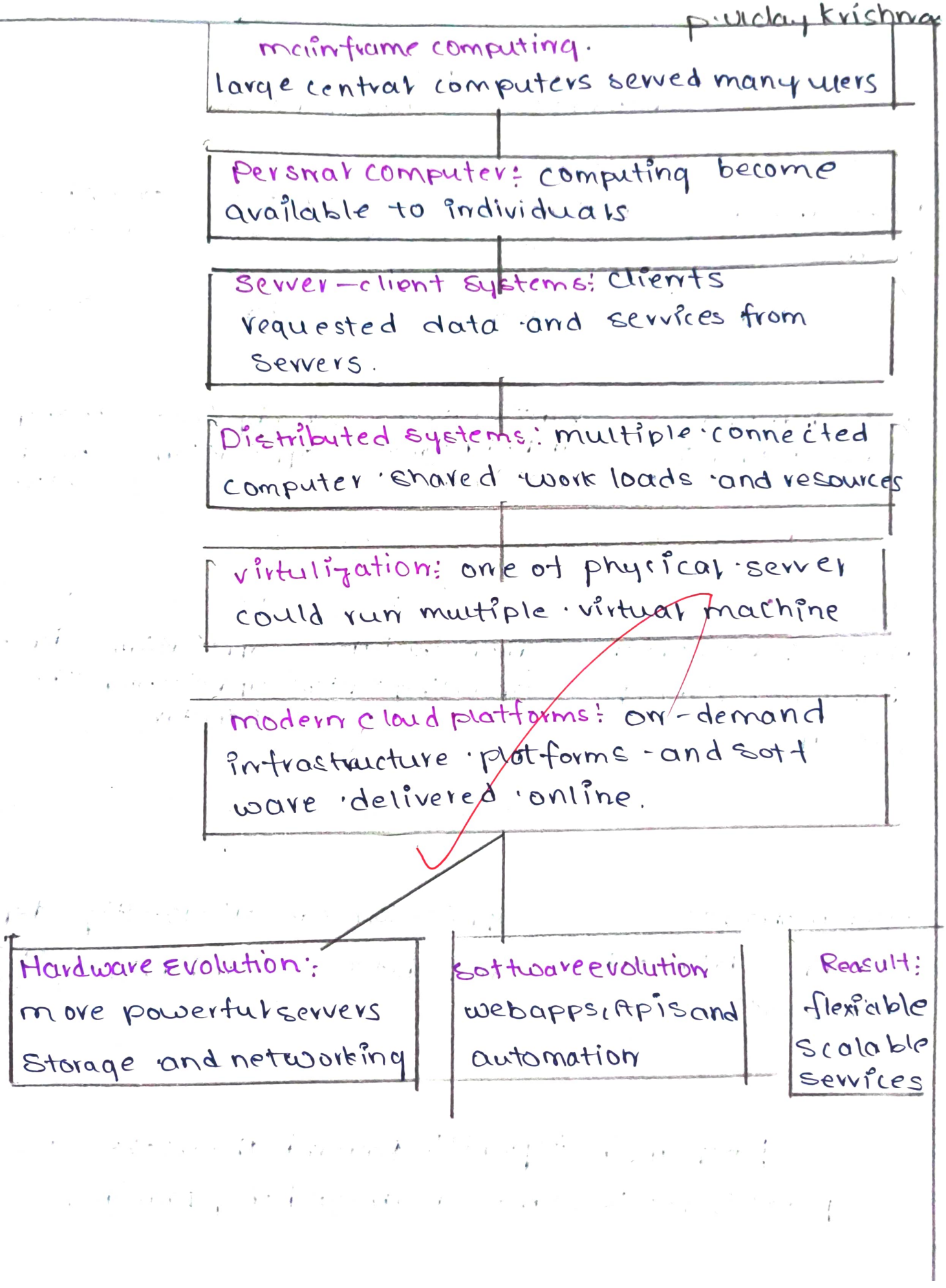
Includes.

measured service
usage is monitered customs may pay for what they we.

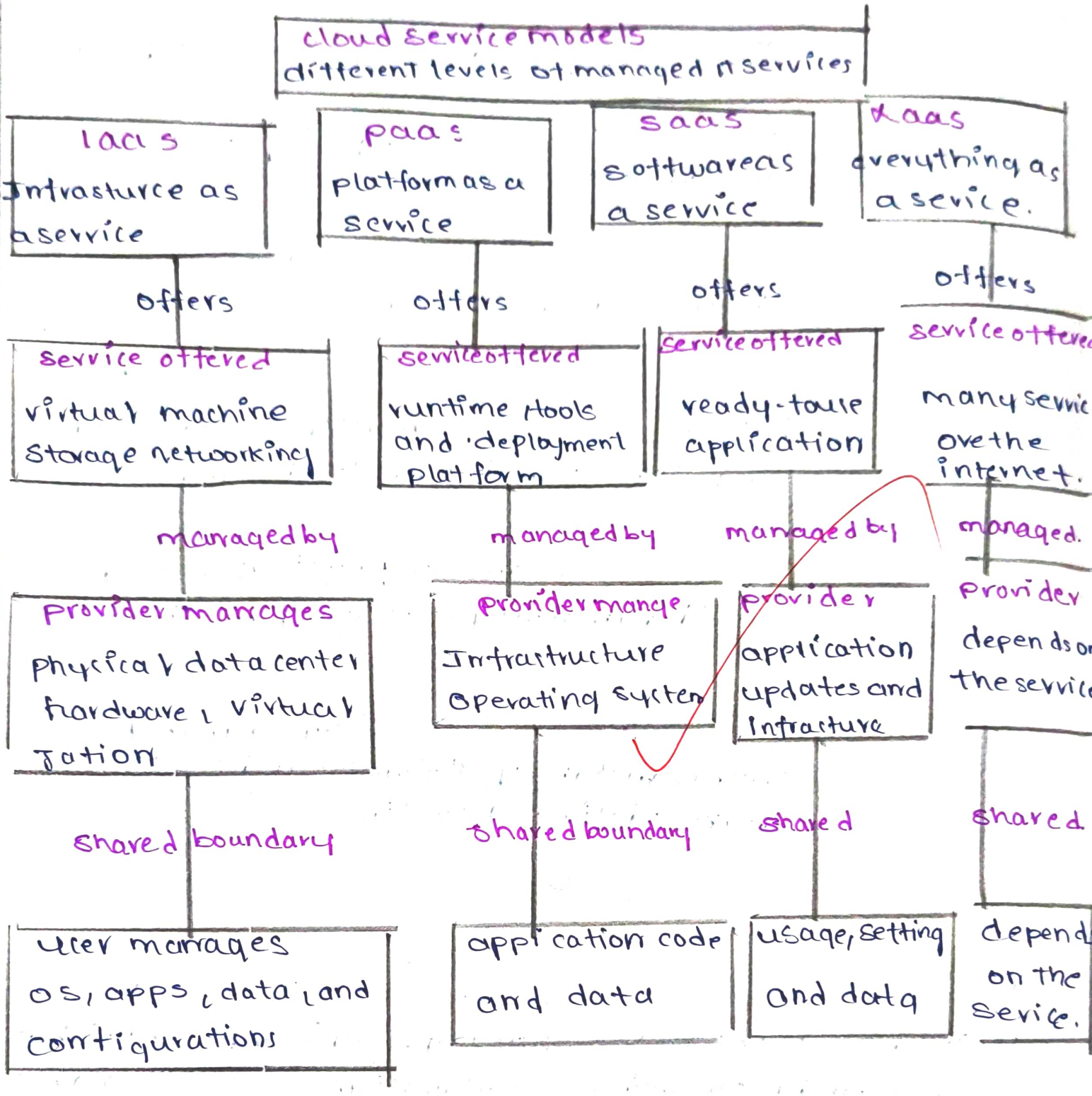
Benefits: lower upfront coit access anywhere quict setup scability. lesshard ware maintanence

ximitations; need internet privacy/security corceins passible down time provider depence ongoing tces

2.


mainframe computing.
large central computers served many users
Personal computer: computing become available to individuals
Server-client systems: clients requested data and services from servers.
Distributed systems: multiple connected computer shared work loads and resources
virtuligation: one of physical server could run multiple virtual machine
modern cloud platforms: on-demand infrastructure plat-forms and soft ware delivered online.
Hardware Evolution:
more powerful servers
Storage and networking
Software evolution
webapps, Apis and automation
Reasult:
flexiable
Scalable
services

3.



Responsibility shifts towards the user ← IaaS/ paas/saas → more managed by provider.

P. Udaykrishna

4.

cloud development + scalability
cloud services are developed, connected and adjusted to

virtuligation
Abstracts physical hardware into virtual sevices

webservices
Allow applications to communitate over a network

uses

creates

Hypervisor
software that create and manages vms

virtual machine
Isolated - Software based computer with os and apps

API
Defined inter for software communication

SOAP
protocal used xml messages.

supports

allows

serves

or

Abstraction
Seperates workhards from specific physical hardsoare.

Resource polling
shares cpu, memory and storage acess

consumer
Apporclient requests a service

Rest
Arch style commonly uling HTTP

support

Improves

request

communication

developmen t
provider allocate compute
Concumer access through
Api of web apps

scalability + claiticit
Add / move virtual resourc
and adjust capacity
with work load

leads to

leads to

Reacult: Efficient resource uce, connected
Sevices and flexible cloud applications

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